# Digital-to-Analog Conversion

ECE 2564, Embedded Systems

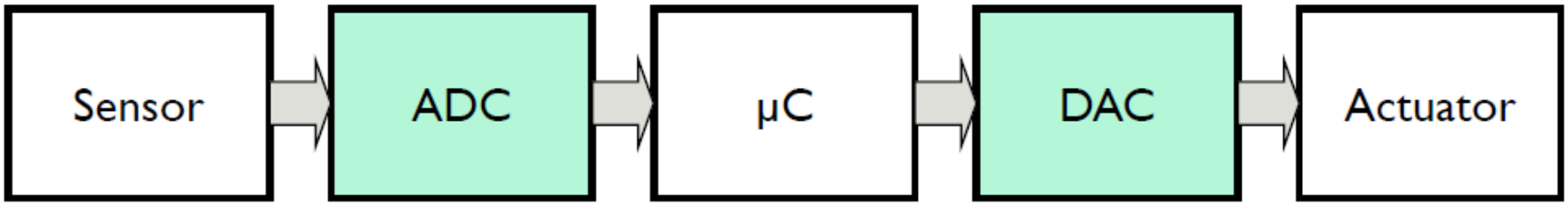
Fall 2025

# Outline

- Quick review (or intro) to op-amp circuits
- Using a Binary Weighted Ladder
- Using an R-2R Ladder

# Digital-to-Analog Conversion

- Goal: convert an unsigned binary number, usually having 8 to 16 bits, to an analog voltage, usually having a common range as $0$ to $V_{CC}$ or $+12$ to $-12\,\mathrm{V}$
- Digital-to-analog conversion normally uses a linear mapping
- The key concerns when performing the conversion are
  - *resolution*, *accuracy* and *speed*



# Digital-to-Analog Conversion

- The conversion is described here:

$$V_{\text{out}} = \frac{D}{2^n}(V_H - V_L) + V_L$$

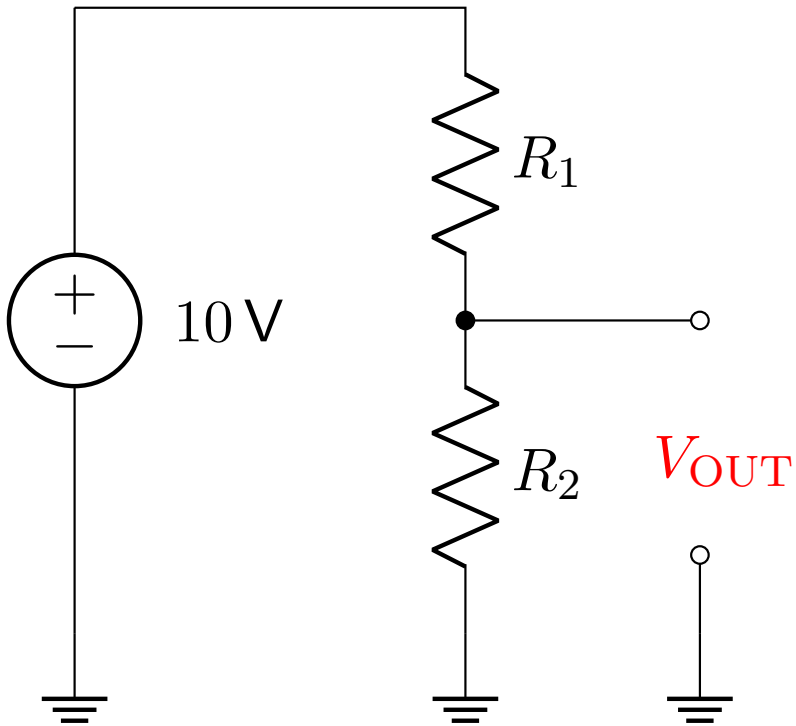
- $D$ is an unsigned digital value to be converted to analog form
- $V_{\text{out}}$ is the *analog output voltage* result of the conversion
- $V_H$ and $V_L$ are the *high reference* and *low reference* voltages
- $n$ is the number of bits that are input to the converter
- Span is the total analog voltage range of the output, $V_H - V_L$
  - Sometimes called the Effective Full-Scale Range, EFSR
- Voltage *resolution*: the span divided by $2^n$
  - the voltage step corresponding to the least significant bit (LSB)

# Basic Idea and Circuit Analysis Review

$$\begin{aligned} 1011_2 &= 1(2^3) + 0(2^2) + 1(2^1) + 1(2^0) \\ &= 1(8) + 0(4) + 1(2) + 1(1) \\ &= 8(1 + 0/2 + 1/4 + 1/8) \end{aligned}$$

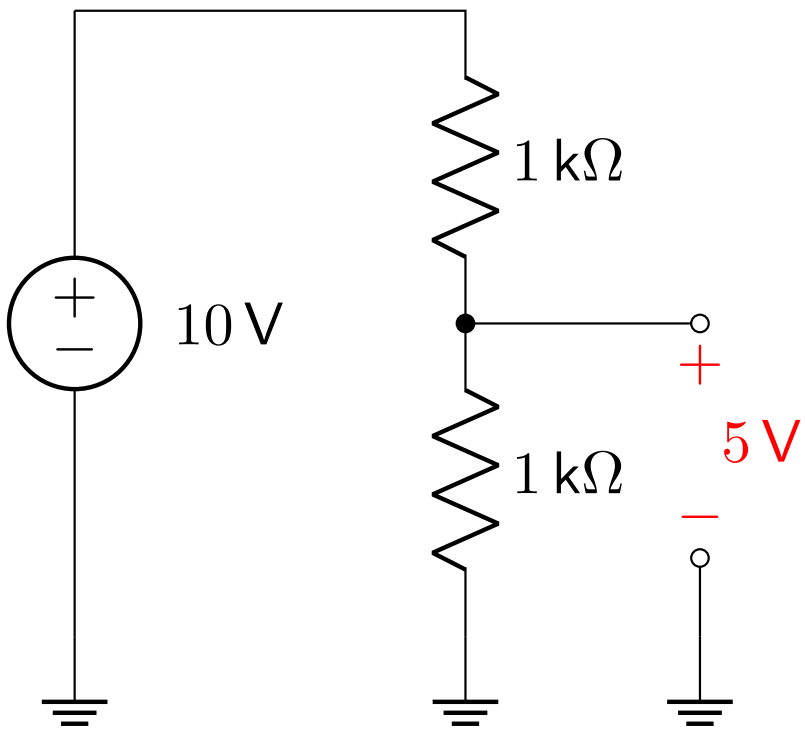
- **Ohm's Law** states that the current through a conductor between two points is directly proportional to the voltage across the two points (multiplication/division)
- **Kirchoff's Current Law**, or KCL, states that the total current or charge entering a junction or node is exactly equal to the charge leaving the node (addition)

# Voltage Dividers



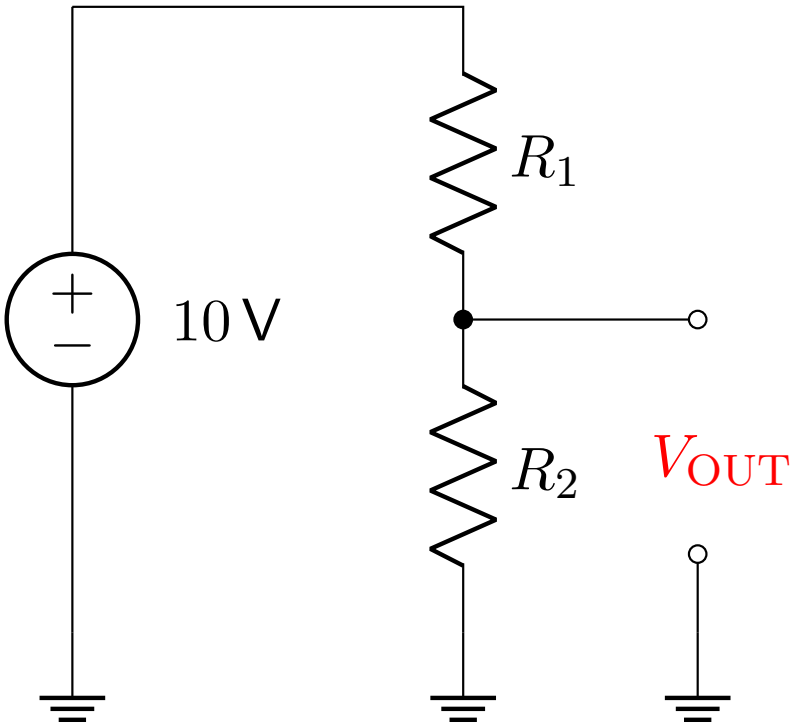

A voltage divider

# Voltage Dividers

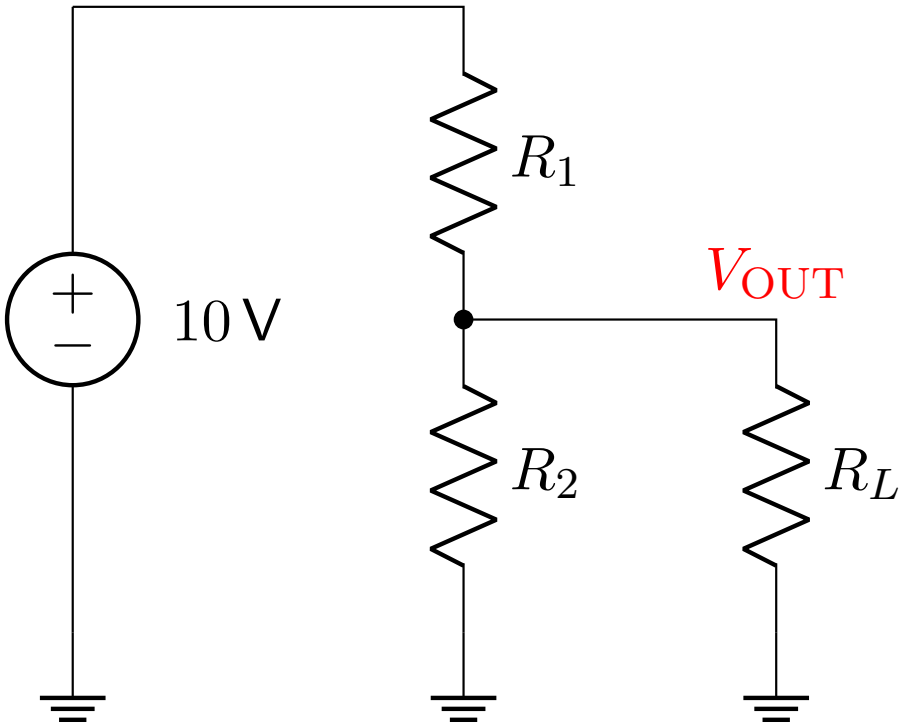



A voltage divider

# Voltage Dividers with Resistive Loads

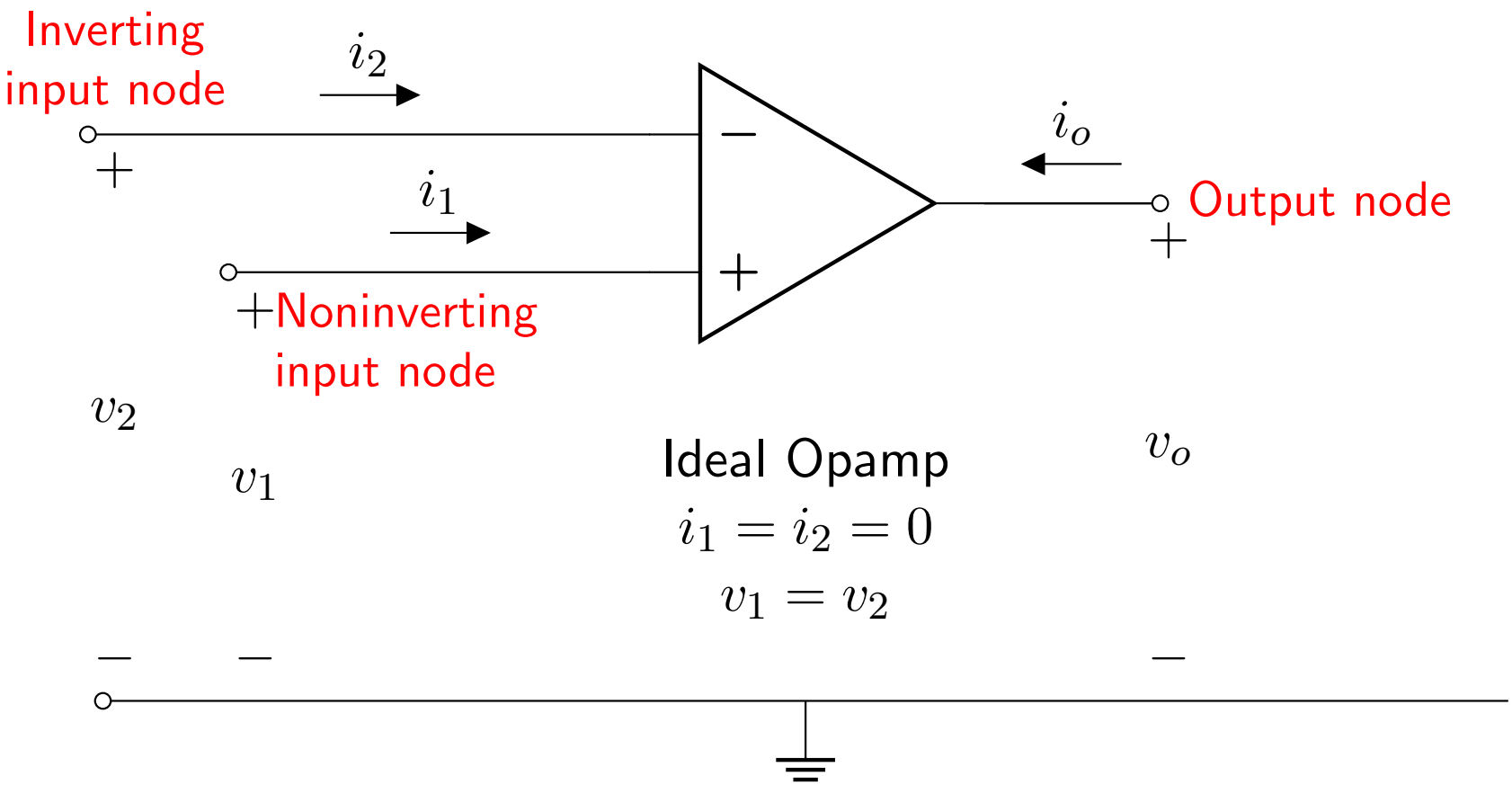



Unloaded Output Voltage

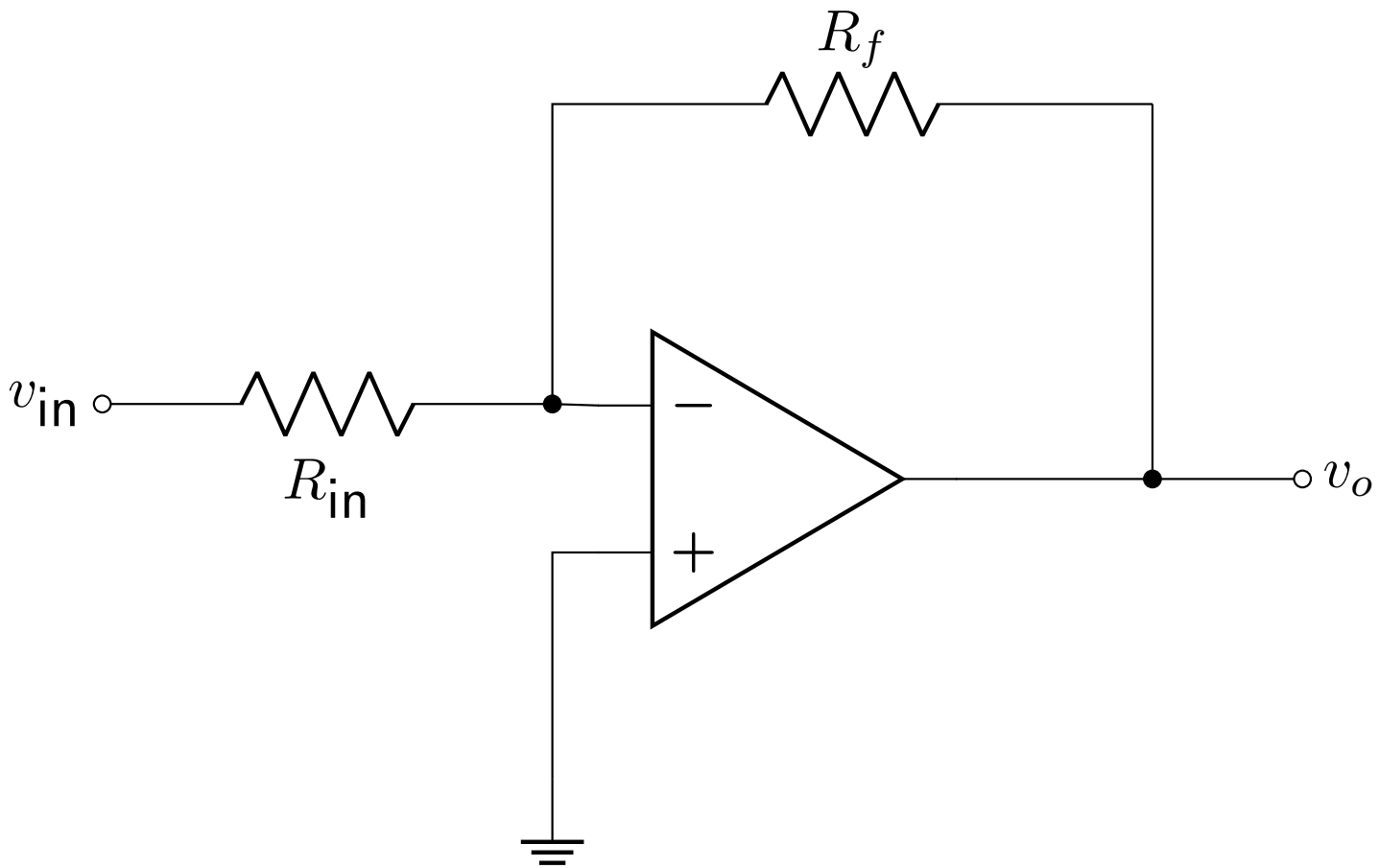



Loaded Output Voltage

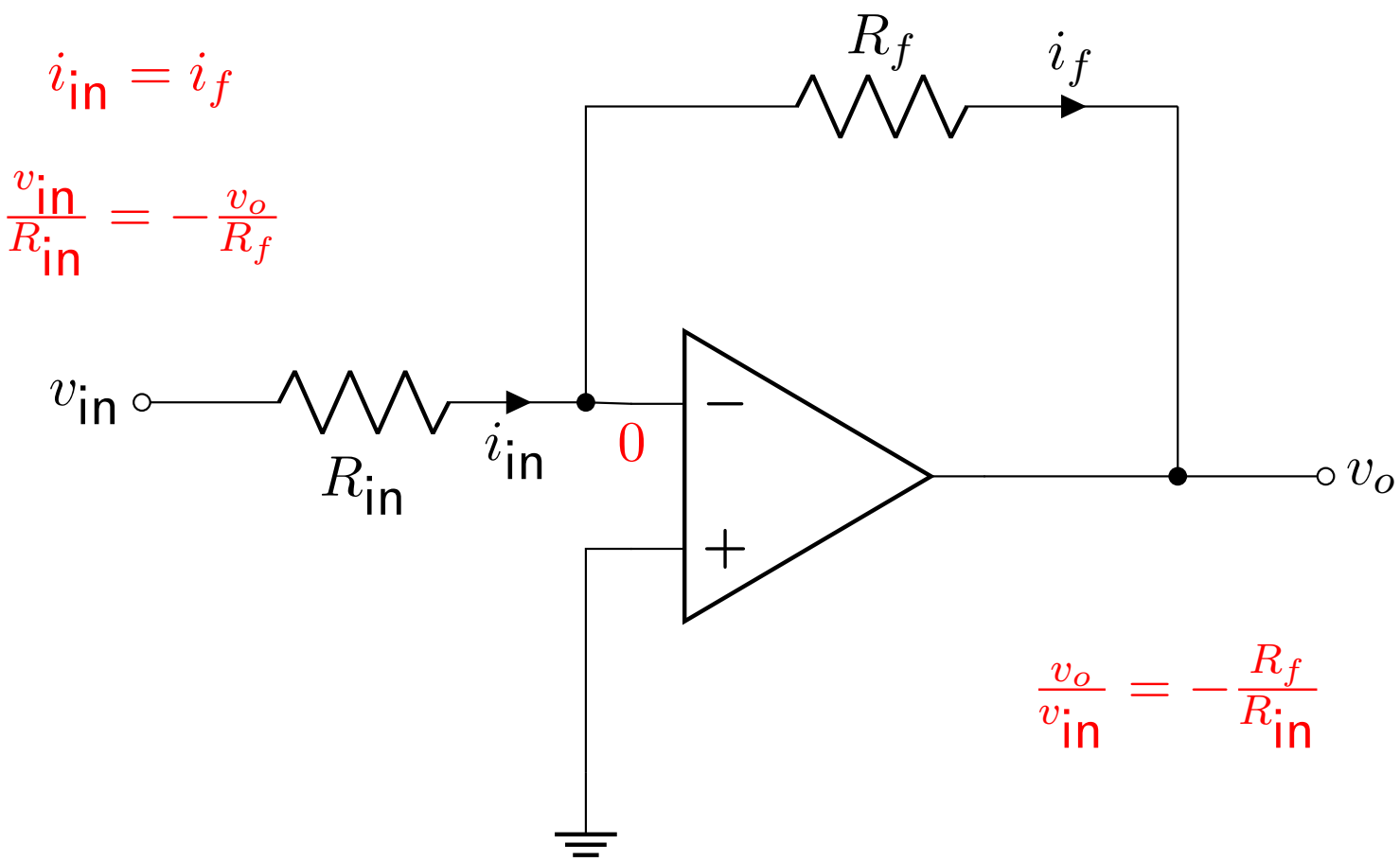
# The Ideal Operational Amplifier



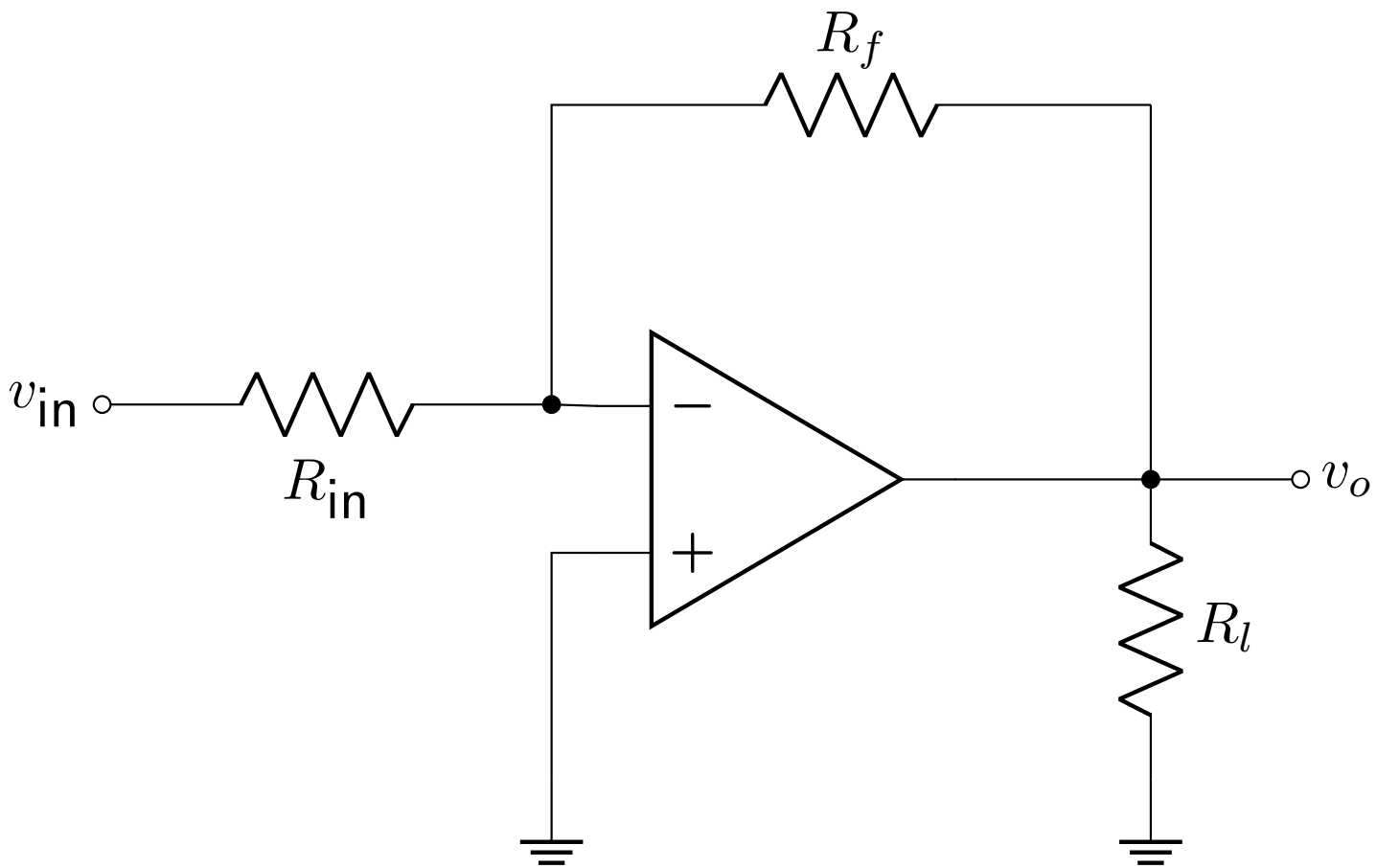
# Inverting Operational Amplifier



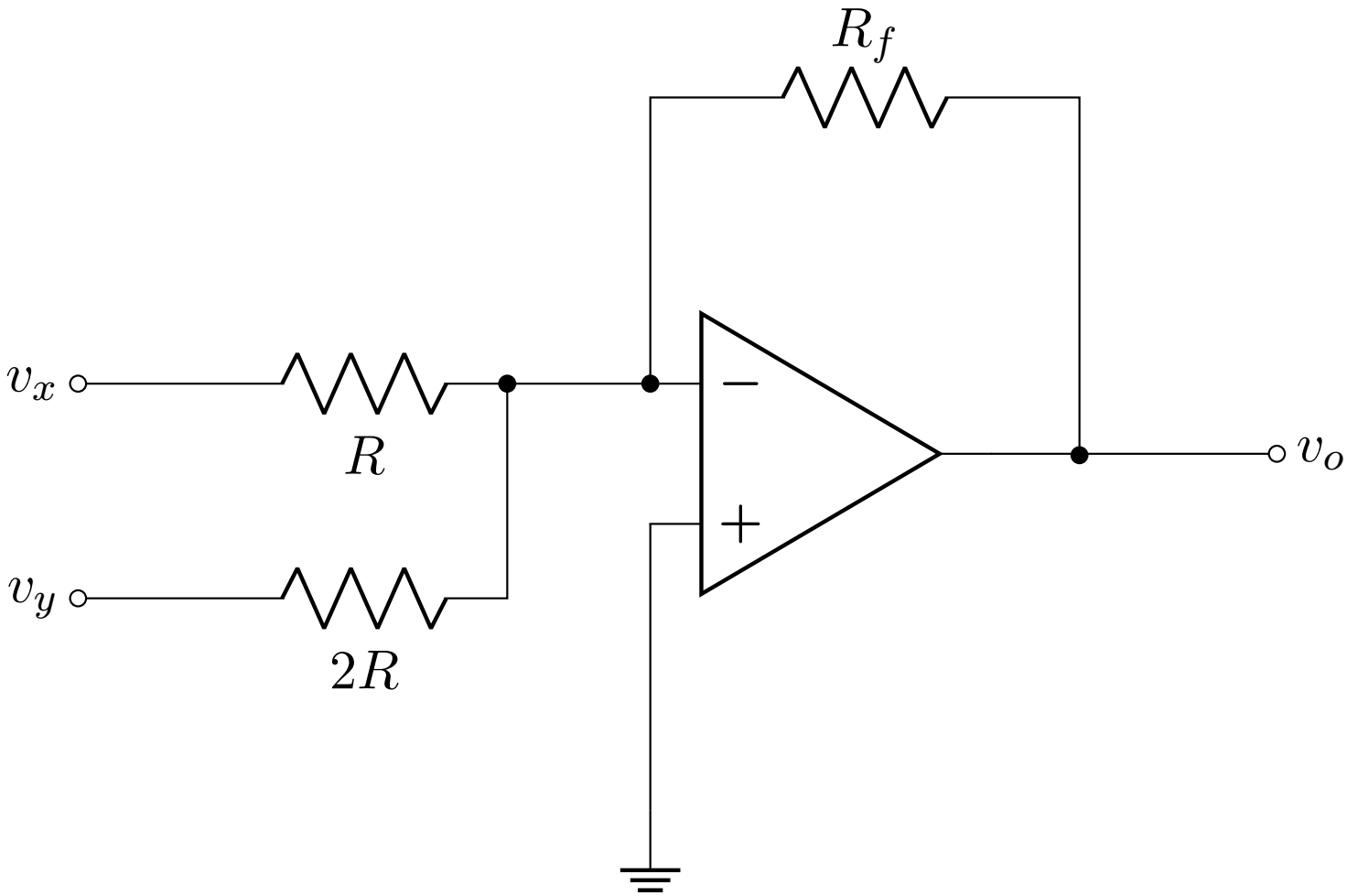
# Inverting Operational Amplifier



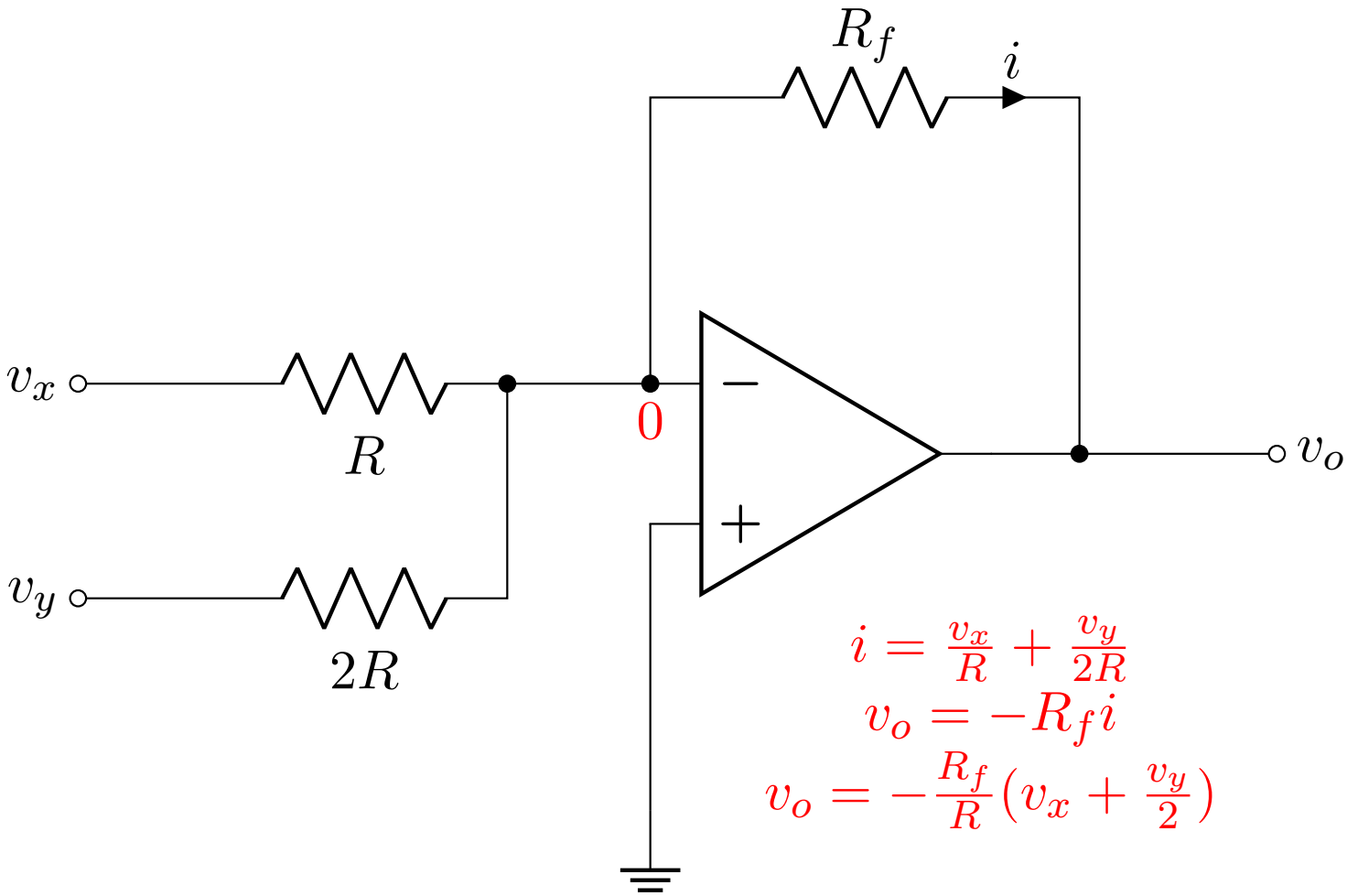
# Inverting Operational Amplifier with Load



# Review of Op-amp Circuits



# Review of Op-amp Circuits




$$i = \frac{v_x}{R} + \frac{v_y}{2R}$$

$$v_o = -R_f i$$

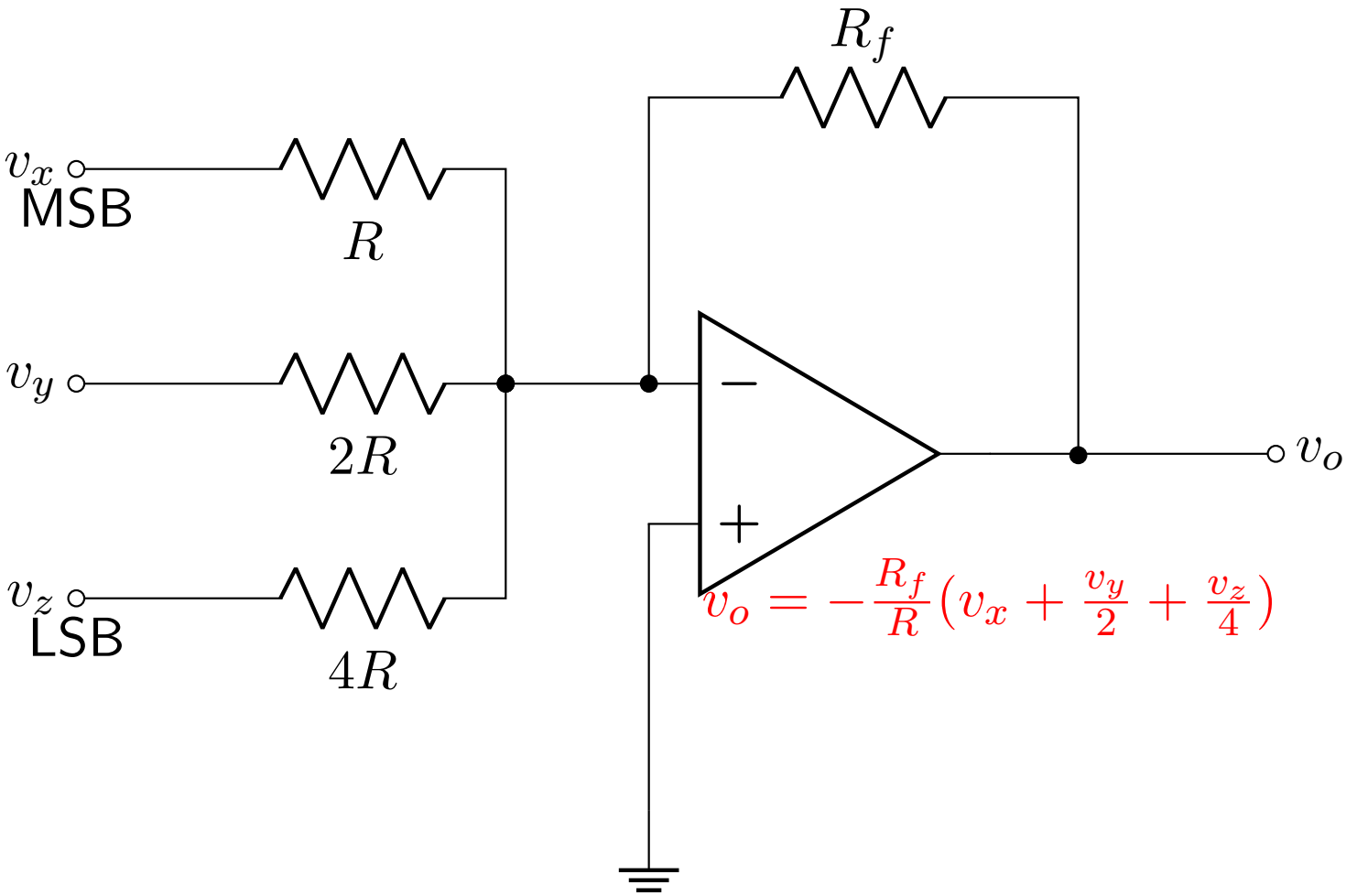
$$v_o = -\frac{R_f}{R}\left(v_x + \frac{v_y}{2}\right)$$

# Our Op-amp Circuit with Binary Inputs

| $v_x$ | $v_y$ | $v_o$ |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | -0.5 |
| 1 | 0 | -1 |
| 1 | 1 | -1.5 |

- Which is the most significant bit (MSB)? $v_x$ or $v_y$ ?
- The MSB has the lowest resistance

# Binary Weighted Ladder – Summing Amplifier



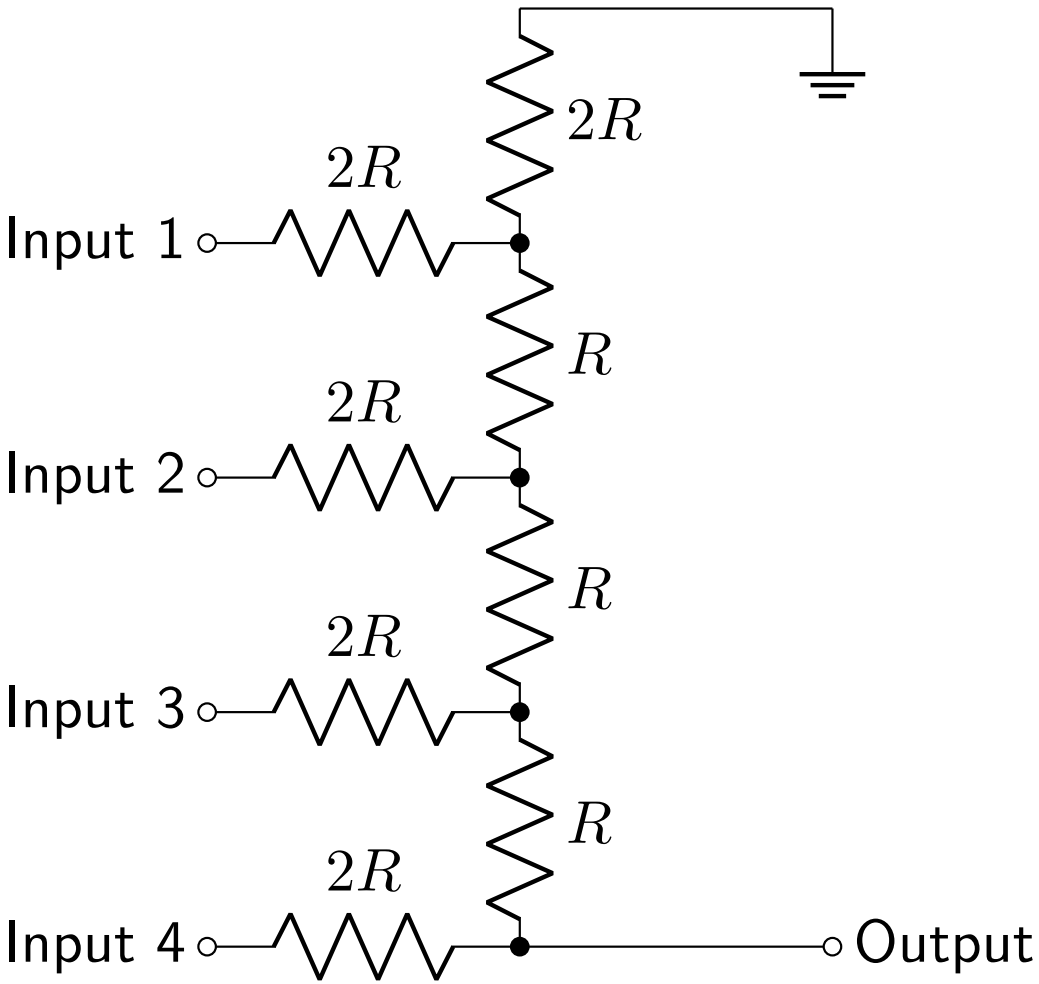
# Binary Weighted Ladder

- Uses one input per bit
  - If the input bit is 1, $V = V_H$
  - If the input bit is 0, $V = 0$
- The value of each input resistor is *twice* that of the previous one and contributes half as much voltage to the sum
- Choose $R_f$ and $R$ to achieve the desired gain

# Binary Weighted Ladder

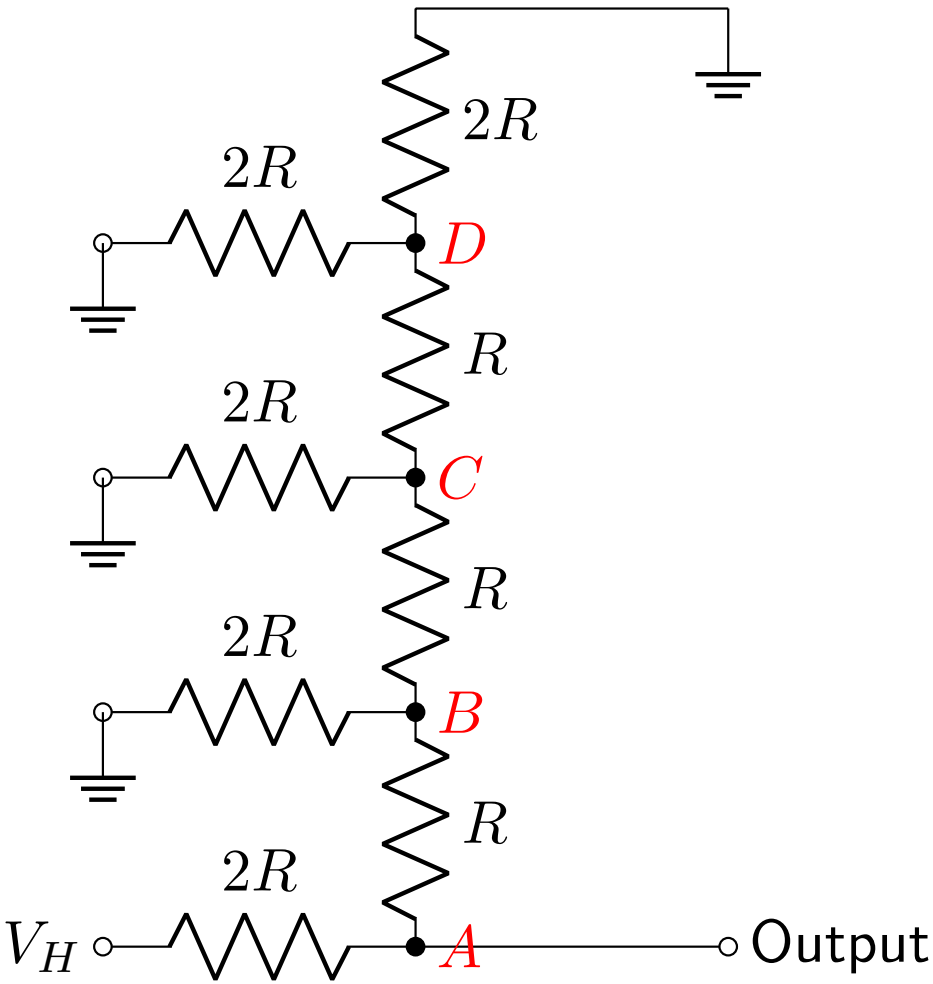
- Fast conversion, but ...
- Not practical for large $n$ because many different *high accuracy* resistors are needed
- The lowest-value resistor $R$ affects the MSB and must have the highest accuracy
  - If the MSB resistor is $1\,\mathrm{k}\Omega$, what is the resistor of the LSB if $n = 10$?
    - $512\,\mathrm{k}\Omega$
- Only useful in case of limited resolution
  - e.g. 8 bits or fewer

# The R/2R Ladder Network




This ladder network is used where digital codes are converted to analog signals, an example of *digital-to-analog* $(D/A)$ convertors. The input signals are the voltage supply, $V_H$ or GND $(0\,\text{V})$.

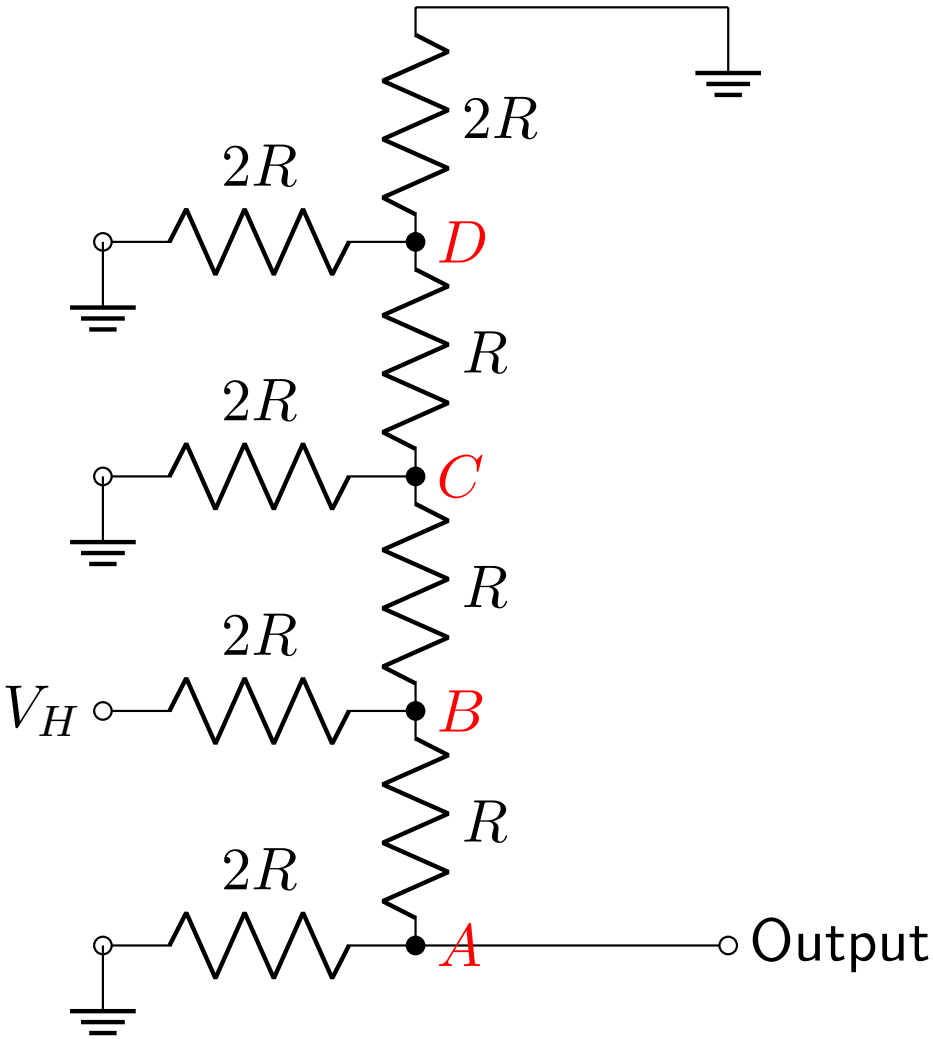
# R/2R Ladder: Example with Input 4 "On"




What is $V_{\text{Output}}$, i.e., $V_A$?

- What is the resistance from $D$ to ground? $2R||2R = R$
- What is the resistance from $C$ to ground? $2R||(R + R) = R$
- What is the resistance from $B$ to ground? $2R||(R + R) = R$
- What is the resistance from $A$ to ground? $R + R = 2R$
- What is the voltage at $A$? $\frac{2R}{2R+2R}V = \frac{V_H}{2}$

# R/2R Ladder: Example with Input 3 "On"



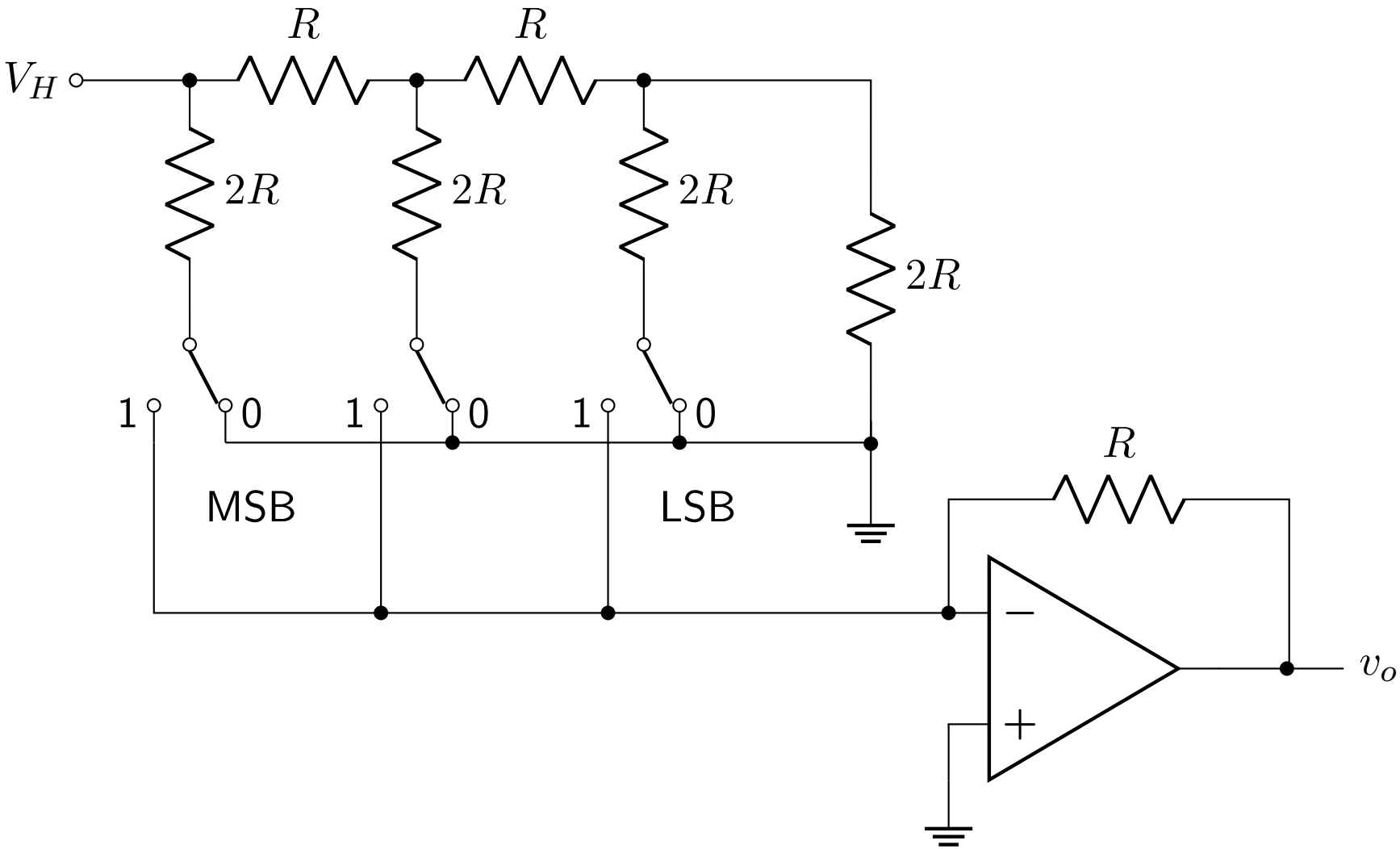

What is $V_{\text{Output}}$, i.e., $V_A$?

- What is the resistance from $D$ to ground? $2R||2R = R$
- What is the resistance from $C$ to ground? $2R||(R+R) = R$
- What is the resistance from $B$ to ground? $2R||(2R+R) = \frac{6}{5}R$
- What is the voltage at $B$? $\frac{\frac{6}{5}R}{\frac{6}{5}R+2R} = \frac{3}{8}V$
- What is the voltage at $A$? $\frac{2R}{3R}\frac{3}{8}V_H = \frac{V_H}{4}$

# R/2R Ladder: Continued

- If Input 2 is "On" and the other inputs are "Off", then $V_{\text{Output}} = \frac{V_H}{8}$
- If Input 1 is "On" and the other inputs are "Off", then $V_{\text{Output}} = \frac{V_H}{16}$
- If more than one input are "On" at a time, the output is the sum of the individual outputs (apply Superposition Theorem)
- The output of the R/2R Ladder is an analog representation of the digital input

# R-2R Ladder DAC



# R-2R Ladder for DAC

- Based on a summing amplifier but it only uses two resistance values, instead of $n$ resistance values
- Using fewer resistance values helps accuracy
- This runs slower than a binary weighted ladded because of the RC/RL effects increase for each addition R-2R stage

# Summary

- Digital-to-Analog conversion methods
  - Using a Binary Weighted Ladder
  - Using an $R - 2R$ Ladder

- Surprise! DAC hardware is sometimes used to build ADC hardware (stay tuned)